

The Effects of High Frequency Traders in a Simulated Market

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Abstract

High frequency trading (HFT) is the latest revolutionary force in financial markets, but good empirical data is notoriously difficult to obtain and analyze. To provide some basic insights into the effects of HFT on stock markets, this paper employs an agent-based simulation to examine the influence of a simple HFT strategy in a continuous double auction market. In the simulated market environment, market liquidity, efficiency, and total surplus each varied directly with the number of traders employing the HFT strategy, thereby strengthening the arguments of some proponents of HFT. However, results supported that market volatility increased with the number of HF traders, and the profits of those traders may come at the expense of long-term oriented investors. These findings may justify some of the concerns of regulators. The simulation setting helps isolate the effect of HFT, and the results bear further testing and validation in empirical markets.

The Effects of High Frequency Traders in a Simulated Market

The last decade has seen a revolution in market microstructure with the advent of high frequency trading (HFT). Hedge funds and proprietary trading desks at major investment banks have invested heavily in technology and human capital to win the race for ever faster trading. While only a few years ago the average NYSE trade took ten seconds to execute (Hendershott and Moulton, 2007), HFT is today driven by powerful computers that employ specialized algorithms to find profitable trading opportunities and execute them in fractions of a second. The practice has been both lucrative and controversial.

HF traders themselves argue that their participation in the market improves markets. Some research findings have illustrated these positive influences through increased liquidity (Hendershott & Riordan, 2009; Jovanovic & Menvel, 2010) and improved price discovery (Brogaard, 2010). Research also provides some evidence of a weak negative correlation between volatility and HFT in the foreign exchange market (Chaboud, Hjalmarsson, Vega, and Chiquoine, 2009).

Opponents claim that HF traders engage in front running and other behavior along the border between unethical and illegal, whether by front running, quote stuffing, or generally creating a two-tiered market (Duhigg, 2009). Additionally, the liquidity argument in favor of HFT is potentially problematic in that HF traders have no affirmative obligation to provide liquidity and might withdraw from the market precisely when liquidity is most needed. Furthermore, some studies suggest that HFT actually increases stock price volatility (e.g., Zhang, 2010). In one extreme example of this, HFT has been blamed for contributing significantly to the “flash crash” of May 6, 2010. In light of these possible abuses, Congress and the SEC have considered a range of potential rules and fees that would potentially limit or penalize HFT.

The dearth of accurate data regarding HFT makes it difficult, if not impossible, to test the various claims of proponents or opponents with direct observation. Partly for that reason but also to provide a controlled environment, this investigation utilizes a market simulation to aid in understanding the influence HFT may have on market dynamics. By eliminating the range of factors that can influence real markets, this experimental market can clarify the relationship among the variables and describe the effect of HF traders on market performance. The next section reviews the relevant literature regarding both high frequency trading and simulated markets, culminating in the hypotheses and research questions this study will address. Following that is the model description and experimental design, the results of the simulation, and a concluding discussion.

Literature Review

The dominant paradigm in economic theory rests on utility maximization of a fully rational representative individual. However, in recent years the discipline has explored the effects of bounded rationality (e.g., Simon, 1991) and heterogeneous behavior of individuals (e.g., Schelling, 1978). The challenge is to understand how such empirical constraints can still give rise to market efficiency and buttress basic economic theory. An early breakthrough in this respect was Becker's (1962) proof that agents who behave randomly within a budget constraint still create downward sloping demand and upward sloping supply curves. By obviating the need for the theoretical assumption of fully rational, utility optimizing individuals, this line of research recognizes that performance of a financial market is a joint result of institutional structure and heterogeneous individual behavior.

In particular, the market structure of a continuous double auction has been recognized for its efficiency in both theory and practice. Such a market allows traders at any time (continuous)

to submit either buy or sell orders (double). Any price that arrives in the market but is not able to be matched is recorded as a limit order in an order book, and waits to be filled by an offsetting bid. Each particular market has its own rules regarding order priority and cancellation, but the basic structure remains unchanged. The availability of public market prices, along with private information, is sufficient to produce efficient resource allocation, an effect that has been termed the Hayek hypothesis (Smith, 1982). Further, Plott (1982) noted that even a market with very few traders will converge to equilibrium if trading takes place in a continuous double auction. In general, this work moves the field away from a Walrasian auctioneer (e.g., Smith, 1962) and more accurately reflects current market practices in most of the major world markets (Friedman, 1984).

By eliminating the human factor, simulated markets allow research to isolate the effect of market structure and basic agent behavior. Following the same evolutionary path as the theory outlined above, simulated markets generally began with a clearinghouse model before moving toward an explicit order book in a continuous double auction setting (Raberto & Cincotti, 2005). The state of the art in this field of agent-based computational finance is summarized by Tesfatsion (2003). While markets are complex, dynamic systems, agent-based financial models typically strip the market down to its basic elements in order to understand a particular issue within markets. Such simple order-driven markets are able to reflect complex phenomena of real stock markets, including excess kurtosis and time-varying volatility (Chiarella & Iori, 2002).

LeBaron (2000) provides a good overview of the simulation literature, from simple single-agent single-asset models to the complex, evolutionary Santa Fe stock market. LeBaron (2001, 2005) also encourages the use of simple models with relatively few parameters and provides general guidance in designing a basic trading model. Parsimony of design overcomes

criticisms that more complex mathematical modeling represents an ad hoc approach in which the assumptions drive the results (Raberto, Cincotti, Focardi, & Marchesi, 2001).

The present study builds principally on the model of Gode and Sunder (1993). Their work is built on zero intelligence traders, who form random reservation prices and interact in a continuous double auction market setting. These traders do not learn or adapt their behavior, nor do they develop trading strategies. However, in a slight adaptation, they are constrained to trade profitably, based on their randomly generated reservation prices. Gode and Sunder (1993) reach five conclusions from their model, but the primary outcome of interest here is that market structure alone is adequate to generate surplus from trading, even in the face of zero intelligence behavior from traders.

Several authors have extended the work of Gode and Sunder (1993) by modifying the behavioral assumptions of the traders. One restriction that can be imposed is that bids and offers are not fully random but some function of recently observed prices. This alteration improves the realism of the simulated market (Jamal & Sunder, 1996; Cliff & Bruten, 1997). In another modification, LiCalzi, Milone, and Pellizzari (2009) altered the assumption of clearing the order book with each completed trade and showed that allocative efficiency is possible under a range of parameter values. Research has also shown that simulations with zero-intelligence traders can exhibit asset price bubbles and crashes (Duffy & Unver, 2006). The present study expands the growing literature of simulated markets with zero-intelligence traders by introducing HF traders.

We now turn to the literature regarding the influence and effect of HFT on various market measures. The first measure of interest is market liquidity, which in this simulated market is synonymous with volume. The role of HF traders can be seen as an evolutionary step from that of a market specialist on the NYSE (Durbin, 2010). Like specialists, HF traders make

profits in part by earning a spread on executed trades, which implies that their trading activity has a direct influence on volume and liquidity in the market. Along those same lines, Chordia, Roll, and Subrahmanyam (2007) claim that higher turnover is associated with more frequent, smaller orders as well as with the increased use of quantitative trading strategies, which typify the activity of HF traders. In a theoretical model, Cvitanic and Kirilenko (2010) show that the increased order flow from HFT increases liquidity. Employing a five-year panel of data from the NYSE, Hendershott, Jones, and Menkveld (2011) find that algorithmic trading causally improves liquidity. In light of this theoretical and empirical research, hypothesis one advances that:

H1. Liquidity (defined as trading volume) will vary directly with the number of HF traders in the simulation.

The efficiency of a market can be measured and defined in various ways. One important goal of financial markets is allocative efficiency, moving resources to their best use.

Informational efficiency in asset pricing and economic efficiency in operations are other vital components of well-functioning markets. In this study, the term efficiency encompasses the degree to which traders are able to realize the potential surplus available from trading. This definition is closely linked to the economic concept of buyer and seller surplus and discovery of the equilibrium, market-clearing price.

Plott and Smith (1978) formalize this concept in defining efficiency as the ratio of actual market surplus to potential market surplus. Potential surplus is the maximum profit for buyers and sellers if all trades take place at the equilibrium price. Put another way, potential surplus is the area between the supply and demand curves to the left of their intersection. By this definition, Gode and Sunder's (1993) model approaches full efficiency in some settings. Hurwicz, Radner, and Reiter (1975) proved that with sufficient levels of retrading, zero-

intelligence market participants can achieve and even exceed one hundred percent efficiency. Therefore, the introduction of HF traders should improve efficiency, leading to the following hypothesis:

H2. Efficiency (the ratio of actual surplus to maximum surplus) will vary directly with the number of HF traders.

The particular market structure employed in this study requires that HF traders carry no overnight inventory. Because of this constraint, increased competition among HF traders or excessive markups might lead to losses at the end of the trading day. This particular expectation is due not to earlier theoretical or empirical work but is an artifact of the simulation structure. Still, it allows a better understanding of the effect of this particular rule, which might be relaxed or altered in future studies. This simulation study explores evidence for the following two hypotheses:

H3. Open positions for HF traders at the end of the trading day will vary directly with the number of HF traders.

H4. Because open positions will sometimes be sold at a loss at the market's closing price, both an increased number of HF traders and higher markups will lead to lower surplus for HF traders.

The debate around HFT is sometimes dominated by opponents who claim that HF traders create froth or excess volatility in the market. There is widespread agreement that HF traders earn higher profits in more volatile markets and therefore trade more heavily in stocks with high volatility (O'Hara, 1995). However, the literature does not always even agree on a proper measure of volatility. Most work defines it as some standard deviation-based concept. However, this raises the issue of sampling frequency and whether the sampling is homogeneous

or nonhomogeneous (Dacorogna et al., 2001). In a novel definition of volatility, Jarnecic and Snape (2010) employ the natural logarithm of the ratio of the daily high and low trading prices. This study uses the most typical and intuitive measure: the standard deviation of the time series of reported trading prices. This time series may be considered non-homogeneous, but consistency across the simulation parameters will at least allow for comparison.

Beyond the definitional issue, researchers have come to different conclusions regarding the influence of HFT on volatility. Brogaard (2010) argues that HFT actually dampens volatility. Similarly, Hendershott and Riordan (2009) find no evidence of increased volatility due to algorithmic trading for firms listed on the Deutsche Bourse and included in the DAX index. An examination of the foreign exchange market similarly reveals no causal relationship between HFT and volatility (Chaboud, Hjalmarsson, Vega, & Chiquoine, 2009), and a similar result has been found using NASDAQ data (Hasbrouck & Saar, 2010). In fact, some models demonstrate that basic algorithmic trading strategies can actually reduce volatility in market prices (Gsell, 2008).

By contrast, some empirical work demonstrates a possible link between HFT and volatility (Arnuk & Saluzzi, 2009b). A positive correlation between price volatility and HFT has been demonstrated even after controlling for exogenous sources of volatility, including firm fundamentals (Zhang, 2010). In theoretical work, Alfarano, Lux, and Wagner (2008) build an analytically tractable model of the foreign exchange market that exhibits excess volatility. Jarrow and Protter (2011) also build a model that, despite omitting a bid-ask spread and assuming perfect liquidity, still exhibits increased volatility following the entrance of HF traders. If HFT were even partially responsible for the “flash crash” of May 6, 2010, this concern for

volatility would be justified (Kirilenko, Kyle, Samadi, & Tuzun, 2011). Given the ongoing debate, research question one addresses the following through simulation output data:

RQ1. What is the effect of HF traders on market volatility (measured as the standard deviation of the price series)?

HF traders are also accused of earning their profits at the expense of long-term investors. This has been of particular concern for regulators and Congress as they strive to maintain fair and orderly markets. If HF traders are passive suppliers of liquidity, then those traders should earn the spread as the cost of transacting, thereby providing a service to other market participants in the form of immediate order fulfillment (Demsetz, 1968; Chacko, Jurek, & Stafford, 2008). However, beyond this zero-sum cost, Arnuk and Saluzzi (2008, 2009a) claim that profitable HFT strategies actually raise transaction costs for other investors. Along the same lines, Roberts (2007) refers to some HFT strategies as sniper algorithms that target profitable opportunities through faster discovery and trade execution. Thus, such trading behavior is potentially a transfer of wealth from liquidity buyers and sellers to HF traders. Examining this issue through simulation provides data from a controlled experiment for comparison to empirical findings.

Therefore, research question two explores:

RQ2. What is the effect of HF traders on market surplus and its components, buyer surplus and seller surplus?

Model Description

The simulation is conducted in the NetLogo environment and builds on the model created by McBride to implement the Gode and Sunder (1993) simulation¹. Following this previous work, three simplifying assumptions are made. First, traders bid and trade only one unit at a time

¹ Information and code for McBride's original implementation of the Gode and Sunder (1993) model can be found at this site: <http://mcbridme.sba.muohio.edu/ace/labs/zitrade/zitradenetlogo.html>. Adapted code to incorporate HF traders is available from the author.

of a single security. Second, a completed transaction cancels any unaccepted bids and offers. Third, when prices cross, the transaction is completed at the earlier quoted price. These three rules govern the trading that takes place in an otherwise typical continuous double auction setting.

To begin each simulated trading day, 50 buyers and 50 sellers are created and assigned a random reservation price between zero and 200. Traders are constrained to require a profit in their trading behavior. Therefore, for the seller this is a minimum acceptable price, while for the buyer the reservation price is a maximum trading price. These 100 liquidity traders are long-term investors who are either taking a position or exiting the market. Because of this, they are limited to one trade over the course of the simulation.

HF traders are distinguished by their repeated trading. At the start of the simulation, they behave like liquidity buyers; they are created with a maximum trading price and enter the pool of potential traders. When a HF trader participates in a trade, however, he immediately calculates a new reservation price by marking up the transaction price by the fixed markup and becomes a potential seller. This very simple strategy is an adaptation from the zero intelligence paradigm, but the strategy remains perfectly algorithmic. HF traders do not adapt or learn from market conditions.

There is no maximum to the number of round-trip trades that a HF trader can complete during the simulation, but HF traders are required to have no inventory at the end of the day. This is consistent with the belief that HF traders act as fast-trading arbitrageurs. Therefore, any remaining shares held by HF traders must be sold at the last reported trading price, which proxies for a market closing price.

Each simulated trading day consists of 2,000 ticks or potential trades. At each tick, a trader is randomly selected from the pool of all traders (buyers, sellers, and HF traders). That trader attempts to buy or sell on favorable terms with another trader in the marketplace. However, if the trader is not immediately able to complete a trade, he enters a bid or offer in the order book to await further traders. At the next tick, another randomly selected trader follows the same algorithm. When a trade is completed, the order book is cleared, and the process continues for 2,000 ticks.

The simulation gathers information on eight dependent variables to test the hypotheses and research questions. Buyer surplus is defined as any profit for liquidity buyers, which is to say any time that buyers are able to purchase a share for less than their reservation price. Seller surplus is defined analogously as profit earned by sellers who are able to complete a trade at a price that exceeds their costs.

Meanwhile, surplus for HF traders is the sum of the surplus earned from both buying and selling behavior. One important difference for HF traders, however, is the possibility of negative surplus for trades at the end of the simulation. Because HF traders are prohibited from holding inventory overnight, they will in some cases need to sell a share at the market's closing price, which may not exceed their purchase price plus markup. The number of open positions that require liquidation at the closing price is another output from the model. Furthermore, in addition to the three sub-categories of buyer, seller, and HF trader surplus, the sum of these three is also calculated as a measure of total market surplus.

The simulation also collects the market's volume, defined as the number of completed trades, and creates a time series of the trading prices from which market volatility is defined as the standard deviation of the series. Finally, efficiency is defined as the ratio of actual market

surplus to potential surplus, following the definition of Plott and Smith (1978). It is a measurement of how much of the available surplus is extracted by market participants. Once the traders are created, the supply and demand curves imply the maximum possible market surplus if all possible trades can be completed within the tick limit and the price constraints. This is then compared to the actual performance of traders in the simulation. Due to the repetitive trading of HF traders, market efficiency can exceed one hundred percent.

The hypotheses and research questions require manipulation of two independent factors: the number of HF traders and their markup percentage. The simulation employs a full factorial design of these two factors. The simulation is run with four different levels for the markup of the HF traders (1, 2, 5, and 10%), while the number of HF traders is set to five different levels: 10, 30, 50, 100, and 200. Because the number of liquidity buyers and sellers is fixed at 50 each in all simulations, these levels corresponds to a HF trader percentage of 9, 23, 33, 50, and 67%, respectively. Each experiment is replicated ten times, giving a total of 200 runs of the simulation. Analysis of the simulation output is tested in eight separate two-way ANOVA tests. For the significant main effects, follow-up pair-wise tests are conducted using Tukey's Honestly Significant Difference (HSD) test.

Results

The results of the two-way ANOVA tests are presented in Table 1. In no case were interaction effects statistically significant, allowing us in the remainder of this section to focus on main effects of the two manipulated variables: number of HF traders and markup. The means of each cell (the average of the ten replications in each case) are presented in Table 2.

The first hypothesis predicted that trading volume would increase with the number of HF traders. The research supports this hypothesis ($F = 82.23$, $df = 4$, $p < 0.001$). The average

volume with 10 HF traders is 31.5, increasing to 42.5 for 200 HF traders. Pairwise tests show that all pairs are significantly different at the 5% level with the exception of 100 and 200 HF traders. It would appear that the increase is not linear but decreases in magnitude as the number of HF traders increases, and that there comes a point where further HF traders have no effect on market volume.

The second hypothesis predicted that an increased presence of HF traders would improve market efficiency. Again, this hypothesis is borne out in the data ($F = 87.49$, $df = 4$, $p < 0.001$). Also as before, all pairs differ at the 5% level except the average outcome with 100 and 200 HF traders. The average efficiency in the presence of HF traders always exceeds 100%, due to the increased number of completed trades. The mean efficiency ranges from 101.8% with 10 HF traders to 147.3% with 200 HF traders in the market.

Because of increased competition from other HF traders, hypothesis three stated that open positions at the end of the trading day would increase with the number of HF traders. This prediction is supported by the data ($F = 140.16$, $df = 4$, $p < 0.001$). Open positions increase from an average of 3.7 with 10 HF traders to 14.1 with 200 HF traders. Tukey's HSD tests show that all pairs differ significantly at the 5% level other than the comparison between markets with 100 and 200 HF traders.

As anticipated in hypothesis four, HF trader surplus differs significantly due both to the number of HF traders ($F = 40.26$, $df = 4$, $p < 0.001$) and markup ($F = 6.48$, $df = 3$, $p < 0.001$). With regard to the number of traders, the surplus for HF traders peaks at 100, but neither the difference between 50 and 200 nor the difference between 100 and 200 is significant. It appears that too much competition among HF traders limits their surplus. Meanwhile, HF trader surplus declines as the markup increases, with the difference between 1 and 10% as well as the

difference between 1 and 20% being statistically significant. As in real financial markets, HF traders do best when they trade repeatedly at low margins.

The effect of HF trading on market volatility is the most controversial and poorly understood aspect of the market. Consequently, we were unable to predict the existence or direction of a relationship *a priori*. Results of research question one show that the number of HF traders has a significant and positive effect on market volatility in the simulation ($F = 12.93$, $df = 4$, $p < 0.001$). Not all pairwise comparisons are significant, however. The means for 10 and 30 traders do not differ, nor do the means for 30 and 50, nor any of the pairwise comparisons of 50, 100, and 200. All other pairs are significant at the 5% level, and the interesting result is that the presence of additional HF traders generally increases price volatility in the market, in contrast to some empirical findings and the claims of HF defenders.

Research question two explores the effect of HF trading on total market surplus and the surplus of liquidity traders. In the simulation, total surplus in the market increases dramatically in the presence of additional HF traders ($F = 41.94$, $df = 4$, $p < 0.001$). However, surplus for liquidity buyers decreases ($F = 41.32$, $df = 4$, $p < 0.001$), while that for sellers actually increases ($F = 88.05$, $df = 4$, $p < 0.001$). The differential effect between buyers and sellers may be an artifact of the simulation structure, in particular the rule that HF traders must close out their positions at the end of the day. The HF traders directly compete with liquidity buyers at the start of every trading day but do not necessarily compete with liquidity sellers because they might simply close out their inventory at the market's closing price.

Discussion

The simulated market in this study validated each hypothesis advanced. First, the participation of HF traders was positive in terms of market liquidity. The simulation model

defined liquidity in a straightforward way as volume in the market, setting aside other considerations of price impact or market depth. Thus, the result that the participation of HF traders increases liquidity is intuitive, and it also coincides with previous research (e.g., Hendershott & Riordan, 2009; Jovanovic & Menkveld, 2010; Cvitanic & Kirilenko, 2010; Hendershott, Jones, & Menkveld, 2011). Other things equal, increased liquidity is an unalloyed good for markets, allowing greater informational efficiency through trading, lower price impact from trading, lower costs of transacting, and lower cost of capital for firms. If this were the sole result of HFT, there would be no controversy.

The second hypothesis suggested that an increased number of HF traders would lead to higher overall market efficiency. The simulation allowed HF traders to trade an unlimited number of times, and this retrading allowed for an increase in market surplus captured by the participant, as predicted in the model of Hurwicz, Radner, and Reiter (1975), who showed that high enough levels of retrading allow efficiency to exceed one hundred percent. The increase in efficiency, in addition to the increased liquidity of hypothesis one, would further argue for unfettered access to markets for HF traders. In these two areas, their trading activity improves market functioning and represents a Pareto improvement for all participants.

As expected from the structure of the simulation and the simple trading strategy employed, increased competition among HF traders eventually leads to more difficulty in executing the strategy profitably. As predicted in the third hypothesis, this leads to more open inventory positions at the end of the simulation period. Closely related is hypothesis four, which predicted that a larger number of HF traders and higher markups in their trading strategy would lead to lower surplus (profitability) for HF traders. Results of the simulation supported this conjecture, which would imply that competition alone will set a natural limit on the behavior of

HF traders. Again, this would bolster proponents of HFT, who believe that it improves markets and is self-regulating through market competition (e.g., Brogaard, 2010). One must question the effectiveness of a natural limit on HFT behavior when estimates of market share have exceeded 70%. As Zhang (2010) notes, there is no apparent need for HFT to exceed the 50% market share that would provide all of the needed liquidity services to the market. It seems excessive to claim that information flows can account for the excess trading. The ubiquity of HFT demands a better understanding of the motives for its existence, particularly if such motives and trading activity result in negative ramifications for equity markets.

There are results of this simulation that would argue for increased scrutiny and even potential regulation of HFT. In particular, the observed increase in volatility when the market includes HF traders in the simulation is noteworthy because this outcome contradicts some of the aforementioned theoretical work as well as the claims of defenders and practitioners of HFT (Gsell, 2008; Chaboud, Hjalmarsson, Vega, & Chiquoine, 2009; Hendershott & Riordan, 2009; Brogaard, 2010; Hasbrouck & Saar, 2010). Instead, this simulation supports the burgeoning literature that suggests HFT might increase price volatility (Arnuk & Saluzzi, 2009b; Zhang, 2010; Jarrow & Protter, 2011). This question took on new urgency following the flash crash. Because HF traders have no affirmative obligation to provide liquidity, they may withdraw from the market precisely when they are most needed, thereby exposing the falsely inflated level of market liquidity and exacerbating volatility in already turbulent markets. This concern is a principal motive for possible regulation of HFT.

One possible reason for the finding of higher price volatility in the simulated market is the simplistic algorithm assigned to HF traders. In real markets, HF traders tend not only to provide liquidity but also to practice arbitrage-based strategies that exploit mispricings, for

example between futures and cash markets. Such trading should close pricing gaps and improve pricing efficiency (Malkiel, 2009). Further study is needed and more complicated models (both simulated and empirical) are necessary to understand fully the effect of HFT on volatility.

The second research question explored the distribution of market surplus among liquidity buyers and sellers and HF traders. In some ways, HF traders can play a role that is similar to a market maker when they provide liquidity to the market. As such, HF traders should earn the spread for providing this service (Demsetz, 1968; Chacko, Jurek, & Stafford, 2008). Potentially, this simply represents a new competitor for traditional market makers, which could lead to lower profits. This would represent a zero-sum game wealth transfer from previous market makers to HF traders.

However, HFT is not monolithic in its trading strategy, and not all HF traders provide liquidity to the market. It is possible that HF traders actually raise transaction costs for liquidity investors (Arnuk & Saluzzi, 2008, 2009a), with sniper algorithms in particular being a source of concern (Roberts, 2007). The concern is that HFT behavior might represent a wealth transfer away from individual investors and institutions toward trading firms. While such a wealth transfer would not, on its own, be sufficient cause for regulation, investors might perceive equity markets to be a rigged game and withhold or withdraw funds. Such concerns for fairness or confidence in the market have been explored, for example, in models of insider trading (e.g., Bhattacharya & Spiegel, 1991; Leland, 1992). A macroeconomic drop in the level of investment could potentially cause more damage to markets than the benefits provided by HFT. Investor reaction to HFT is an empirical matter that lies outside the scope of this simulation study, but the implication that HF traders might make excess profit at the expense of liquidity traders is troubling and bears further scrutiny.

In recent years, the HFT revolution has quickly come to represent the majority of trading volume. While precise figures are not available, SEC Chairman Mary Schapiro referred to estimates exceeding 50% of total volume in her May 6, 2010, testimony before Congress. Important decisions regarding taxation and regulation will continue to be made, and the public's understanding and confidence in markets will be shaped by the debate and the media coverage. The urgent need to understand the effect of HFT is clear. This simulation study supports the claims of supporters that HFT increases volume and market efficiency. However, the results also raise concerns regarding volatility and wealth transfers that require further study. Maintaining transparency, fairness, and an efficient market microstructure should be the continuing goal. This simulation provides one more data point in exploring that aim.

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Table 1. Two-way ANOVA Results

Source of variation	Sum of squares	df	Mean square	F	p-value
Dependent variable: Volume					
HF Traders	3,129.67	4	782.42	82.23	0.000
Markup	23.81	3	7.94	0.83	0.477
HF traders x Markup	150.41	12	12.53	1.32	0.212
Residual	1,712.70	180	9.52		
Dependent variable: Efficiency					
HF Traders	53,512	4	13,378	87.49	0.000
Markup	2,867	3	956	6.25	0.000
HF traders x Markup	1,626	12	136	0.89	0.562
Residual	27,525	180	153		
Dependent variable: Open HF positions					
HF Traders	2,820.43	4	705.11	140.16	0.000
Markup	13.66	3	4.55	0.9	0.440
HF traders x Markup	78.97	12	6.58	1.31	0.217
Residual	905.50	180	5.03		
Dependent variable: HF trader surplus					
HF Traders	4,151,056	4	1,037,764	40.26	0.000
Markup	501,418	3	167,139	6.48	0.000
HF traders x Markup	359,891	12	29,991	1.16	0.313
Residual	4,639,635	180	25,776		
Dependent variable: Volatility					
HF Traders	414.08	4	103.52	12.93	0.000
Markup	47.72	3	15.91	1.99	0.118
HF traders x Markup	75.69	12	6.31	0.79	0.663
Residual	1,441.31	180	8.01		
Dependent variable: Total market surplus					
HF Traders	27,809,753	4	6,952,438	41.94	0.000
Markup	778,936	3	259,645	1.57	0.199
HF traders x Markup	2,888,435	12	240,703	1.45	0.146
Residual	29,841,369	180	165,785		
Dependent variable: Buyer surplus					
HF Traders	4,605,562	4	1,151,391	41.32	0.000
Markup	19,629	3	6,543	0.23	0.872
HF traders x Markup	253,456	12	21,121	0.76	0.693
Residual	5,016,110	180	27,867		
Dependent variable: Seller surplus					
HF Traders	30,524,900	4	7,631,225	88.05	0.000
Markup	187,981	3	62,660	0.72	0.539
HF traders x Markup	1,685,650	12	140,471	1.62	0.089
Residual	15,600,711	180	86,671		

Table 2. Means of dependent variables

Variable: Volume		HF Traders					Overall
Markup		10	30	50	100	200	
1%		32.6	35.2	39.2	41.3	43.7	38.4
2%		32.5	36.3	39.3	41.2	43.2	38.5
5%		30.2	36.7	37.7	42.8	40.7	37.6
10%		30.7	38.1	39	41.4	42.3	38.3
Overall		31.5	36.6	38.8	41.7	42.5	38.21
Variable: Efficiency		HF Traders					Overall
Markup		10	30	50	100	200	
1%		100.3	123.3	129	150.4	154.7	131.6
2%		106	120.1	130	142.2	155.1	130.7
5%		99.6	118.6	125.7	138.2	139.2	124.2
10%		101.5	114.2	121	138	140.2	123
Overall		101.8	119.1	126.4	142.2	147.3	127.36
Variable: Open HF		HF Traders					Overall
Markup		10	30	50	100	200	
1%		4.2	8.5	11.5	13.3	15.4	10.58
2%		3.7	8.9	11.6	12.8	13.8	10.16
5%		3.4	9.1	10.6	12.7	14.2	10
10%		3.6	9	13	14.6	12.8	10.6
Overall		3.72	8.88	11.67	13.35	14.05	10.34
Variable: HF Surplus		HF Traders					Overall
Markup		10	30	50	100	200	
1%		11	199	287	460	520	295.2
2%		52	261	370	419	381	296.6
5%		-3	168	263	405	334	233.4
10%		4	62	189	398	225	175.6
Overall		16	173	277	420	365	250.22

Table 2. Means of dependent variables

Variable: Volatility		HF Traders					Overall
Markup		10	30	50	100	200	
1%		37.6	39.7	38.9	41.3	41.5	39.8
2%		39.6	40.1	42	40.9	42.7	41.1
5%		37.8	40.3	41.6	42.3	42	40.8
10%		37.8	39.1	40.9	42.6	42.5	40.6
Overall		38.2	39.8	40.9	41.8	42.2	40.56
Variable: Total Surplus		HF Traders					Overall
Markup		10	30	50	100	200	
1%		2576	2784	3044	3573	3918	3179
2%		2658	2758	3086	3338	3624	3093
5%		2466	2931	2984	3536	3251	3034
10%		2477	2922	3000	3246	3462	3021
Overall		2544	2849	3028	3423	3564	3081.71
Variable: Buyer surplus		HF Traders					Overall
Markup		10	30	50	100	200	
1%		1138	956	854	747	728	885
2%		1120	947	888	786	660	880
5%		1048	946	935	789	589	861
10%		1068	1019	947	724	674	886
Overall		1094	967	906	762	663	878.13
Variable: Seller surplus		HF Traders					Overall
Markup		10	30	50	100	200	
1%		1428	1629	1903	2366	2671	1999
2%		1485	1549	1828	2134	2583	1916
5%		1420	1818	1786	2341	2328	1939
10%		1405	1840	1865	2124	2563	1959
Overall		1435	1709	1845	2241	2536	1953.36